DDoS Lab Toolkit for Ethical Hacking Training

Introduction

This document provides a list of tools and methods for **ethical DDoS testing** in a lab environment. All activities should be performed on **owned or authorized networks** only.

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1. LOIC (Low Orbit Ion Cannon)

- **Purpose:** Simulate TCP/UDP floods.
- **Use case:** Beginner-friendly tool for lab experiments.
- **Notes:** Never use on public networks.
- **Example:** Flood a test VM port 80 in a controlled environment.

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2. Hping3

- **Purpose:** Craft custom TCP/UDP/ICMP packets for stress testing.
- **Use case:** Flexible simulation of SYN, UDP, or ICMP floods.
- **Example SYN flood (lab only):**

```
sudo hping3 -S -p 80 --flood <target_ip>
```

- **Notes:** Highly configurable for advanced lab exercises.

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3. Metasploit Auxiliary Flood Modules

- **Purpose:** Simulate service-specific DoS attacks.
- **Use case:** Integrate into pentesting labs to teach flood techniques.
- **Example module:** `auxiliary/dos/tcp/synflood`

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4. HOIC (High Orbit Ion Cannon)

- **Purpose:** Advanced flood simulation for lab VMs.
- **Use case:** Multiple attacker VMs can stress-test a target VM.
- **Notes:** Only in isolated lab setups.

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5. Nmap DoS Scripts

- **Purpose:** Service-specific DoS testing.
- **Use case:** Teach interns about vulnerability-based attacks.

- **Command example:**

  ```
  nmap --script dos -p 80 <target_vm_ip>
  ```

- **Notes:** Not a generic DDoS tool; safe for lab use.

Ethical Guidelines

1. Run all tests in **isolated lab networks**.
2. Use **VMs** to isolate attackers and targets.
3. Monitor system resources; floods can crash your host.
4. Focus on **learning defenses** and attack mechanics, not harming external systems.

Lab Setup Recommendation

- **Attacker VMs:** Kali Linux with LOIC/Hping3/Metasploit.
- **Target VM:** Ubuntu/Windows/Linux with test services (HTTP, FTP, SMTP).
- **Network:** Internal host-only or NATed VM network.
- **Monitoring:** Use tools like `top`, `htop`, or Wireshark to observe impact.

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